To determine your Internet Protocol (IP) address, execute the `ifconfig` command:

```
sysadmin@localhost:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1450
        inet 192.168.1.2  netmask 255.255.255.0  broadcast 0.0.0.0
        inet6 fe80::42:c0ff:fea8:102  prefixlen 64  scopeid 0x20<link>
        ether 02:42:c0:a8:01:02  txqueuelen 0  (Ethernet)
        RX packets 11  bytes 878 (878.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 648 (648.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1  (Local Loopback)
        RX packets 1  bytes 49 (49.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1  bytes 49 (49.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
sysadmin@localhost:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         192.168.1.1     0.0.0.0         UG    0      0        0 eth0
192.168.1.0     0.0.0.0         255.255.255.0   U     0      0        0 eth0
```

To view the table of routing information, use the `route` command:

```
sysadmin@localhost:~$ grep 127.0.0.1 /etc/hosts
127.0.0.1       localhost
```

Verify that the IP address `127.0.0.1` has an entry in the `/etc/hosts` file:

```
sysadmin@localhost:~$ ping -c4 localhost
PING localhost(localhost (::1)) 56 data bytes
64 bytes from localhost (::1): icmp_seq=1 ttl=64 time=0.032 ms
64 bytes from localhost (::1): icmp_seq=2 ttl=64 time=0.035 ms
64 bytes from localhost (::1): icmp_seq=3 ttl=64 time=0.029 ms
64 bytes from localhost (::1): icmp_seq=4 ttl=64 time=0.035 ms

--- localhost ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2997ms
rtt min/avg/max/mdev = 0.029/0.032/0.035/0.007 ms
```

Test to see if the `localhost` machine will respond to four `ping` requests:

```
sysadmin@localhost:~$ cat /etc/resolv.conf
search vixen.lb
nameserver 127.0.0.11
options ndots:0
```

view the `/etc/resolv.conf` file to see if any `nameserver` entries exists:

```
sysadmin@localhost:~$ dig localhost.localdomain

; <<>> DiG 9.11.3-1ubuntu1.5-Ubuntu <<>> localhost.localdomain
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 56461
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: 0708514fa6562be38505ea9d6000c50b4aea1805343782f8 (good)
;; QUESTION SECTION:
;localhost.localdomain.         IN      A

;; ANSWER SECTION:
localhost.localdomain.  86400   IN      A       127.0.0.1

;; AUTHORITY SECTION:
localdomain.            86400   IN      NS      localhost.localdomain.

;; Query time: 1 msec
;; SERVER: 127.0.0.11#53(127.0.0.11)
;; WHEN: Thu Feb 22 02:22:45 UTC 2024
;; MSG SIZE  rcvd: 108                        1
```

Use the `dig` command to resolve the `localhost.localdomain` name to an IP address:

```
sysadmin@localhost:~$ dig cserver.example.com

; <<>> DiG 9.11.3-1ubuntu1.1-Ubuntu <<>> cserver.example.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25402
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: c33d9913b915efc1e055c1575c006fe19ca3975ce1951a1c (good)
;; QUESTION SECTION:
;cserver.example.com.           IN      A

;; ANSWER SECTION:
cserver.example.com.    86400   IN      A       192.168.1.2

;; AUTHORITY SECTION:
example.com.            86400   IN      NS      example.com.

;; ADDITIONAL SECTION:
example.com.            86400   IN      A       192.168.1.2

;; Query time: 0 msec
;; SERVER: 127.0.0.11#53(127.0.0.11)
;; WHEN: Thu Feb 22 02:24:57 UTC 2024
;; MSG SIZE  rcvd: 122
```

use the `dig` command to resolve other fully-qualified domain names. Use the `dig` command to resolve the `cserver.example.com` hostname to an IP address

```
sysadmin@localhost:~$ dig -x 192.168.1.2

; <<>> DiG 9.11.3-1ubuntu1.1-Ubuntu <<>> -x 192.168.1.2
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 59190
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;2.1.168.192.in-addr.arpa.      IN      PTR

;; ANSWER SECTION:
2.1.168.192.in-addr.arpa. 600   IN      PTR     84a6444e-f044-4754-98e6-64cc9f7d
adea.1.84a6444e-f044-4754-98e6-64cc9f7dadea.LAN.

;; Query time: 0 msec
;; SERVER: 127.0.0.11#53(127.0.0.11)
;; WHEN: Thu Feb 22 02:25:29 UTC 2024
;; MSG SIZE  rcvd: 159
```

Use the `dig` command to resolve the IP address `192.168.1.2` to a hostname:

```
sysadmin@localhost:~$ netstat --help
usage: netstat [-vWeenNcCF] [<Af>] -r         netstat {-V|--version|-h|--help}
       netstat [-vWnNcaeol] [<Socket> ...]
       netstat { [-vWeenNac] -i | [-cnNe] -M | -s [-6tuw] }

        -r, --route              display routing table
        -i, --interfaces         display interface table
        -g, --groups             display multicast group memberships
        -s, --statistics         display networking statistics (like SNMP)
        -M, --masquerade         display masqueraded connections

        -v, --verbose            be verbose
        -W, --wide               don't truncate IP addresses
        -n, --numeric            don't resolve names
        --numeric-hosts          don't resolve host names
        --numeric-ports          don't resolve port names
        --numeric-users          don't resolve user names
        -N, --symbolic           resolve hardware names
        -e, --extend             display other/more information
        -p, --programs           display PID/Program name for sockets
        -o, --timers             display timers
        -c, --continuous         continuous listing

        -l, --listening          display listening server sockets
        -a, --all                display all sockets (default: connected)
        -F, --fib                display Forwarding Information Base (default)
        -C, --cache              display routing cache instead of FIB
        -Z, --context            display SELinux security context for sockets

  <Socket>={-t|--tcp} {-u|--udp} {-U|--udplite} {-S|--sctp} {-w|--raw}
           {-x|--unix} --ax25 --ipx --netrom
  <AF>=Use '-6|-4' or '-A <af>' or '--<af>'; default: inet
  List of  possible address families (which support routing):
```

The `netstat` command performs a large variety of tasks related to networking. To get an idea of some of its capabilities, execute the command with the `--help` option

```
sysadmin@localhost:~$ netstat -tl
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:domain        0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp        0      0 localhost:domain        0.0.0.0:*               LISTEN
tcp        0      0 localhost:953           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.11:38453        0.0.0.0:*               LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
tcp6       0      0 [::]:domain             [::]:*                  LISTEN
sysadmin@localhost:~$ netstat -tln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 192.168.1.2:53          0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:53            0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:953           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.11:38453        0.0.0.0:*               LISTEN
tcp6       0      0 :::22                   :::*                    LISTEN
tcp6       0      0 :::53                   :::*                    LISTEN
```

Use the `netstat` command to see if the TCP port 22, for `ssh`, has a process listening:

```
sysadmin@localhost:~$ netstat -ltn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 192.168.1.2:53          0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:53            0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:953           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.11:44441        0.0.0.0:*               LISTEN
tcp6       0      0 :::22                   :::*                    LISTEN
tcp6       0      0 :::53                   :::*                    LISTEN
```

```
sysadmin@localhost:~$ start_webserver
Connecting ...
Connecting ...
Connecting ...
Connecting ...
```